

CLOCK IN · SOLANA MOBILE HACKATHON

Bao 紅包

*Red packets for Seeker.
One Seeker, one grab.*

Drop SKR into your circle. Friends shake their phones to grab a share. Bots are refused by the program itself.



THE PROBLEM

Every crypto giveaway gets farmed.

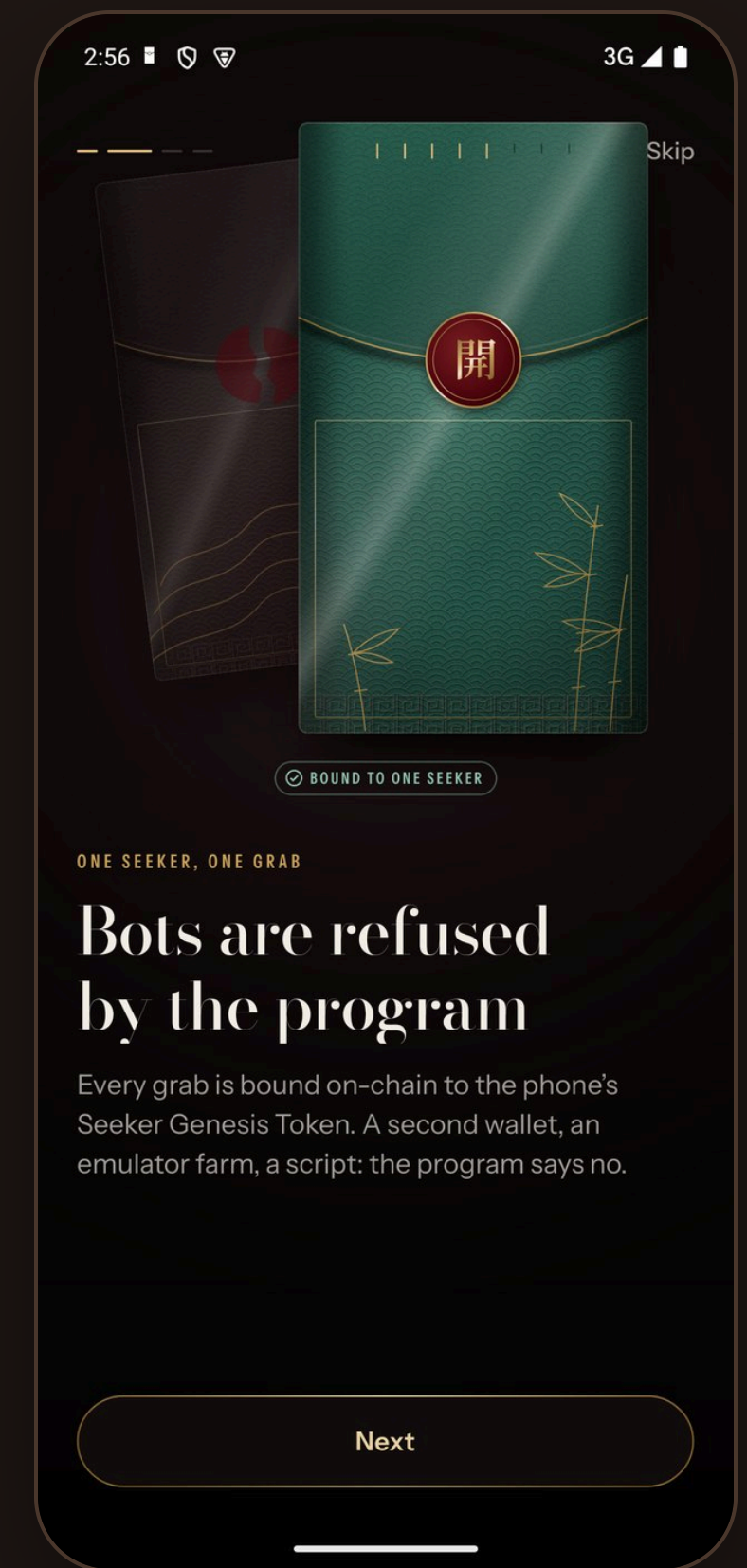
Airdrops, raffles and “first 100 wallets” drops go to whoever runs the most wallets. Projects pay to reach bots, real people get crumbs, and nobody comes back.

THE IDEA

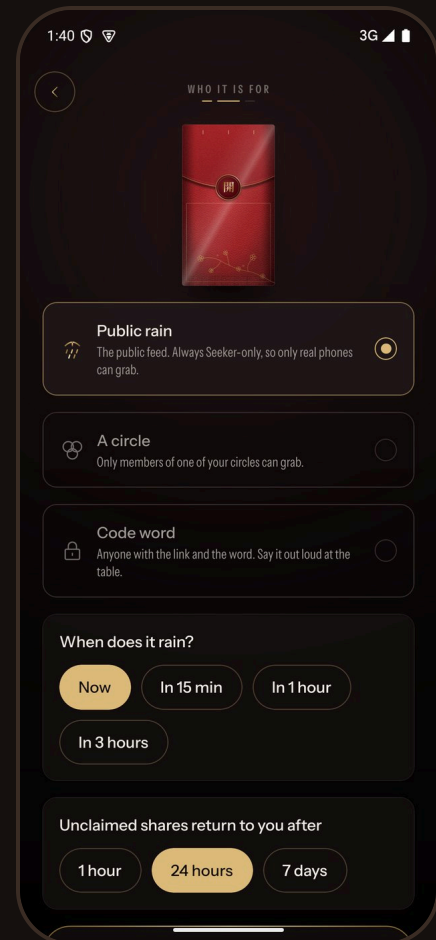
The hongbao, on a phone that can prove it is one phone.

The red packet is how hundreds of millions of people already send money as play. Bao rebuilds it on Solana and binds every grab to the phone's Seeker Genesis Token.

One device, one grab — however many wallets it holds.



How a packet works



1 • Drop

Amount, shares, Lucky or Equal; a circle, the open feed or a code word. One Seed Vault signature.



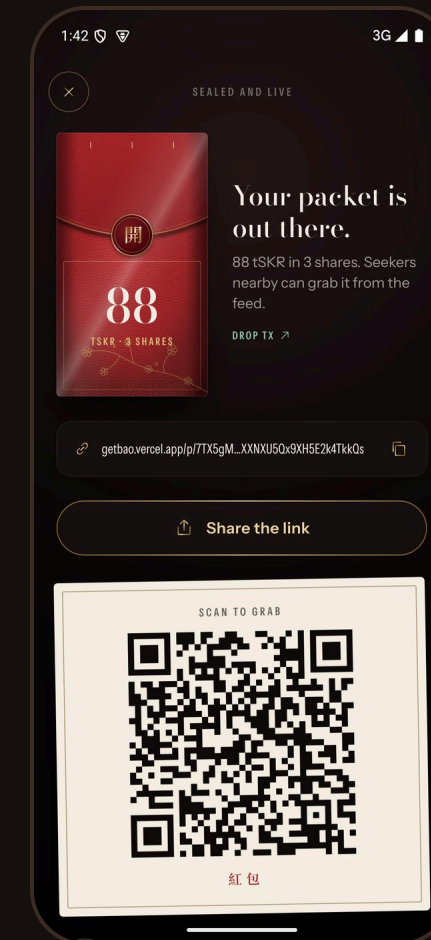
2 • Shake

Three shakes break the seal. The program checks the Genesis Token and reserves a share.



3 • Reveal

MagicBlock VRF proves the random share in about 2.5 s. The biggest grab is crowned Luck King.



4 • Pass it on

The Luck King sends the next one. Share by link, QR or NFC tag; a crank pays out and refunds the rest.

THE REFUSAL IS THE PRODUCT

Bots are refused by the program,
not by a server.

Attempt	The program answers
Wallet without a Seeker Genesis Token	NotASeeker · 6013
Same phone, a second wallet	AlreadyGrabbedOnThisDevice · 6016
Wrong code word	WrongCode · 6012

The claim record is keyed by the Genesis Token’s mint, not the wallet, so moving the token to a fresh wallet does not buy a second grab. Each refusal has a devnet transaction in PROOF.md; the screen on the right is a real phone being refused.



PROVABLE RANDOMNESS

Luck anyone can verify

2.3 s

from grab to a proven random share, measured on devnet with MagicBlock VRF (2.2–2.7 s).

1 proof

per Lucky grab. The callback assigns the share with WeChat's double-mean split; the packet's own gas tank pays the oracle.

0

payouts without a proof. A request the oracle never answers can be cancelled after 300 slots and refunded.

Only makes sense on a Seeker

Genesis Token

The sybil filter, checked on-chain at every grab.

Seed Vault

Signs each drop and grab through Mobile Wallet Adapter.

Shake and haptics

Opening a packet is a physical gesture, with sound.

NFC and QR

Hand packets out in person: tap a tag or scan a code.

Home-screen widget

Live packets and the next rain, without opening the app.

Push

When a packet lands in your circle or your share arrives.

.skr names

Shown in place of addresses in circles and feeds.

*None of it works
for a farm of
emulators.*

SKR INTEGRATION

SKR is the currency of every packet.

Senders spend it

A packet is SKR held in a vault owned by the program until it is grabbed. Gifting gives people a reason to spend SKR, not just hold it.

Grabbers receive it

Shares land as SKR in the grabber's wallet, and the Luck King custom sends it on. On devnet the stand-in is tSKR with the same 6 decimals.

Projects distribute it

Rains put SKR in front of verified Seeker owners only, one share per phone. No staking involved.

Why people come back tomorrow

Packets come from people you know

Circles with invite codes, Luck King chains and leaderboards.

The home screen shows what is waiting

The widget counts live packets and the next rain.

Rains are scheduled

A push when one starts; shaking opens the moment it begins.

The Luck King owes the next packet

By custom the biggest grab sends the next one, so the circle returns.



SUSTAINABILITY

How Bao pays for itself

1%

protocol fee on open packets and rains that reach every verified Seeker. Enforced on-chain; the sender caps it with `max_fee_bps`.

Free

for circles. Friends gifting friends pay Bao nothing, which is what keeps the habit going.

Prepaid

randomness. Each packet funds its own gas tank for the oracle and a reward for the crank that closes it, so grabbers pay only the network fee.

For projects, a rain is sybil-proof distribution to real Seeker owners: the thing airdrops never managed.

Built, tested, live on devnet

76

program tests in LiteSVM, 16 of them adversarial

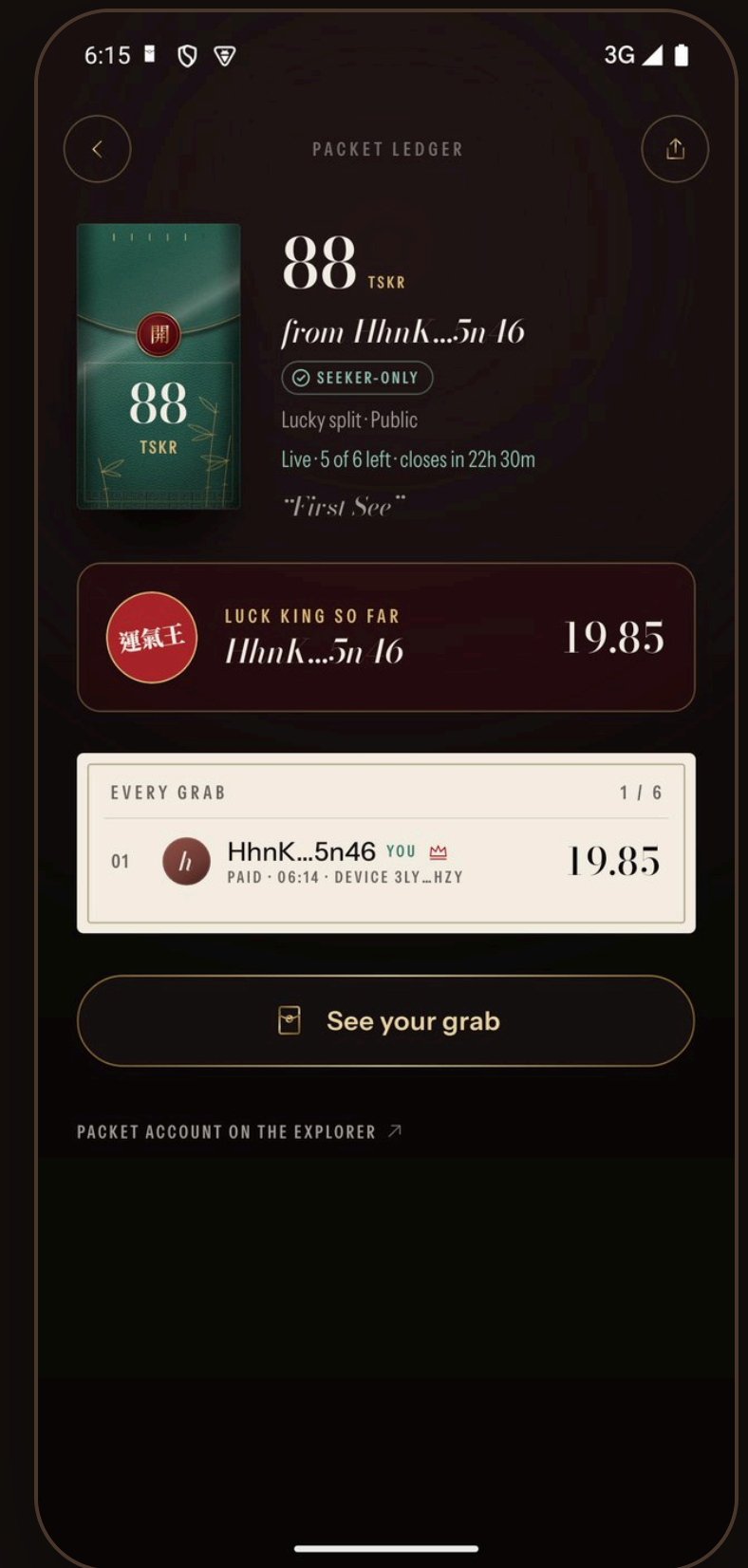
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API tests: crank, indexer, circles, push, actions

21

SDK tests, share math pinned against the program

Anchor program with MagicBlock VRF · Android app on Expo with Mobile Wallet Adapter and Skia · API, crank, indexer and push on Vercel and Supabase · Helius webhooks. A real mainnet Seeker Genesis Token passes the same check in a fixture test. Every transaction and the threat model are in the repository.



WHAT IS IN THIS BUILD

Working, devnet-only, planned. Nothing is mocked.

Feature	Status in the submitted APK
Drop, grab, payout, refund (Anchor program)	Working on devnet
One grab per Seeker Genesis Token (6013, 6016)	Working on devnet with a test group; real mainnet token layout proven in a test
Verifiable randomness per grab (MagicBlock VRF)	Working on devnet
Wallet connect, sign-in and signing (Mobile Wallet Adapter)	Working; tested with Phantom on a Samsung A52, not yet on a Seeker
Shake, haptics, widget, push, QR, App Links, circles, rains	Working
Writing a packet to an NFC tag	Built; not tried on a physical tag
SKR	tSKR stand-in on devnet; real SKR on mainnet is planned
Mainnet, dApp Store, multisig upgrade authority	Planned

On devnet so far: 37 packets, 32 grabs, 30 randomness proofs, 26 payouts, 344 of 1,701 tSKR grabbed. These are the builder’s own test runs on one phone, one emulator and scripted wallets; there are no outside users yet.

What comes next

Step	What it means
Mainnet	The real Seeker Genesis group and real SKR, same program and checks.
dApp Store	Publish the signed APK to the Solana dApp Store.
Hardening	Upgrade authority to a multisig, then frozen; an external audit.
In person	NFC packet drops at Breakpoint London, November 15–17.

Bao 紅包

Red packets bots can't eat.

Try it

Where

Android APK

github.com/RaYYeR220/bao/releases

Demo video

youtu.be/N1hNVH_5Gmo

Code and proof

github.com/RaYYeR220/bao

Live API

getbao.vercel.app